Current: charge flow rate

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate current as charge transferred per time.

SUCCESS CRITERIA - I can...

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “charge” to explain your model or result. Sentence starters are optional.

current: rate of charge flow

charge: a property responsible for electric interactions

interval: the time or range between two stated values

01 Bring back an earlier idea

Both charge magnitudes become 3 times as large and separation becomes 5 times as large. Express new force as a fraction of the old.

Word help: separation = distance between the specified objects or centers

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $I = \Delta q/\Delta t$.
Current is charge per time; 1 A = 1 C/s. Conventional current follows positive charge motion.

Worked reasoning: If 12 C pass a point in 4 s, average current is I=Q/t=3 A. Charge circulates; a resistor does not consume charge.

Your turn: Draw a complete loop and label equal current before and after one resistor in steady state.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw conventional current around the closed circuit, outside the battery from + toward -. What happens if a wire is disconnected?

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Solve and justify

A wire carries .25 A for 8 minutes.
Find charge transferred.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

At a junction, 5 A enters and 2 A leaves along one branch.
Find the other outgoing current in steady operation.
Explain the conservation rule.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A source approaching a stationary listener at 20 m/s emits 600 Hz.
Use $v_{sound}=340$ and $f_{obs}=f_s\ v/(v-v_s)$.
Find observed frequency.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Does a large electron drift speed have to equal the speed at which a switched circuit responds?
Explain qualitatively.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A current rises linearly from 0 to 4 A during 3 s, then stays 4 A for 2 s.
Find total charge from the current-time graph.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3